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WO-400-02 — Overhaul pump P-103B (mechanical seal and bearings)

Area	A400 Pumps	**Task**	PO Pump overhaul
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	P-103B — P-103B centrifugal pump — E 179'-0" N -18'-6" EL 100'-9"		
Equipment	P-103B	**Lines**	4"-P300-P-1014, 3"-P300-P-1016, 3/4"-P150-W-3022
Isometrics	ISO-4"-P300-P-1014, ISO-3"-P300-P-1016, ISO-3/4"-P150-W-3022 (generated); owner's drawings ISO-P-1014, ISO-P-1016, ISO-W-3022		
Permits	General work permit, Lifting plan		

Reason

High bearing vibration; overhaul seal and bearings.

Scope of work

1. Isolate P-103B electrically and per the isolation list; drain the casing.
2. Record as-found shaft alignment; remove the coupling guard and spacer.
3. Break the suction and discharge nozzle joints; remove the pump to the workshop.
4. Replace the mechanical seal and the bearings; inspect the impeller and wear rings.
5. Reinstall; new nozzle gaskets and studs tightened per SW-WO-5; align to the owner's alignment criteria.
6. Remove blinds and isolations; run test.

Affected items: P-103B

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1014	4"-P300-P-1014	4"-P300-P-1014	1	0 error(s), 1 warning(s) — see below
ISO-P-1016	3"-P300-P-1016	3"-P300-P-1016	1	1 error(s), 0 warning(s) — see below
ISO-W-3022	3/4"-P150-W-3022	3/4"-P150-W-3022	1	0 error(s), 1 warning(s) — see below

ISO-P-1014 (PDF): ../isometrics-imported/ISO-P-1014.pdf · ISO-P-1016 (PDF): ../isometrics-imported/ISO-P-1016.pdf · ISO-W-3022 (PDF): ../isometrics-imported/ISO-W-3022.pdf — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1014	1	warning	branch weld 3/4"-P300-I-7035/W1 (E 176'-6" N -11'-0" EL 102'-3 5/16", sheet 1) has no label
ISO-P-1016	1	error	sheet 1: CONT'D ON ISO-P-1015 SH 1 prints E 174'-2 1/2" N -17'-6" EL 107'-6" for E 174'-2 1/2" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 3"-P300-P-1016 meets 3"-P300-P-1015 (E 174'-2 1/2" N -16'-6" EL 107'-6")
ISO-W-3022	1	warning	weld 3/4"-P150-W-3022/W8 (E 181'-0" N -16'-6" EL 124'-10 3/4", sheet 1) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-103B	equipment	P-103B centrifugal pump		E 179'-0" N -18'-6" EL 100'-9"
P-103B-M	motor	Motor of P-103B		E 179'-0" N -18'-6" EL 100'-9"
4"-P300-P-1014/B7	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1014	E 179'-0" N -13'-3-11/16" EL 102'-0"
4"-P300-P-1014/B5	item	FLANGE WELD-NECK 4" P300 RAISED FACE	4"-P300-P-1014	E 179'-0" N -14'-2-1/8" EL 102'-0"
3"-P300-P-1016/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1016	E 179'-0" N -16'-6" EL 105'-2-3/16"
3"-P300-P-1016/A2	item	VALVE, SWING CHECK 3" P300	3"-P300-P-1016	E 179'-0" N -16'-6" EL 104'-1/8"
3/4"-P150-W-3022/A3	item	VALVE, BALL, FLOATING 3/4" P150	3/4"-P150-W-3022	E 178'-0" N -4'-0" EL 123'-7-11/16"
3/4"-P150-W-3022/A6	item	FLANGE WELD-NECK 3/4" P150 RAISED FACE	3/4"-P150-W-3022	E 178'-0" N -4'-0" EL 124'-4-5/8"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	electrical	P-103B-M		Motor of P-103B: isolated, locked and tried at the starter			
2	valve	4"-P300-P-1014/B7	4"-P300-P-1014	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1014/B7) on P-103B-S	E 179'-0" N -13'-3-11/16" EL 102'-0"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
3	blind	4"-P300-P-1014/B5	4"-P300-P-1014	Fit a blind between 4"-P300-P-1014/B7 and 4"-P300-P-1014/B5, P-103B side	E 179'-0" N -14'-4-7/8" EL 102'-0"		
4	valve	3"-P300-P-1016/A5	3"-P300-P-1016	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1016/A5) on P-103B-D	E 179'-0" N -16'-6" EL 105'-2-3/16"		
5	blind	3"-P300-P-1016/A2	3"-P300-P-1016	Fit a blind between 3"-P300-P-1016/A5 and 3"-P300-P-1016/A2, P-103B side	E 179'-0" N -16'-6" EL 103'-5"		
6	valve	3/4"-P150-W-3022/A3	3/4"-P150-W-3022	Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-W-3022/A3) on P-103B-F	E 178'-0" N -4'-0" EL 123'-7-11/16"		
7	blind	3/4"-P150-W-3022/A6	3/4"-P150-W-3022	Fit a blind between 3/4"-P150-W-3022/A3 and 3/4"-P150-W-3022/A6, P-103B side	E 178'-0" N -4'-0" EL 124'-2-3/8"		

Parts

Description	Qty	Replaces
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1014/B1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1014/B3
GASKET 3" P300 COMPRESSED SHEET	1	3"-P300-P-1016/A1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	3"-P300-P-1016/A3
Mechanical seal kit for P-103B	1	
Bearing set for P-103B	1	

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1014/B1	4 × 1" studs (4"-P300-P-1014/B3)	35,343	3	
3"-P300-P-1016/A1	4 × 0.875" studs (3"-P300-P-1016/A3)	27,059	3	

Test — Run test

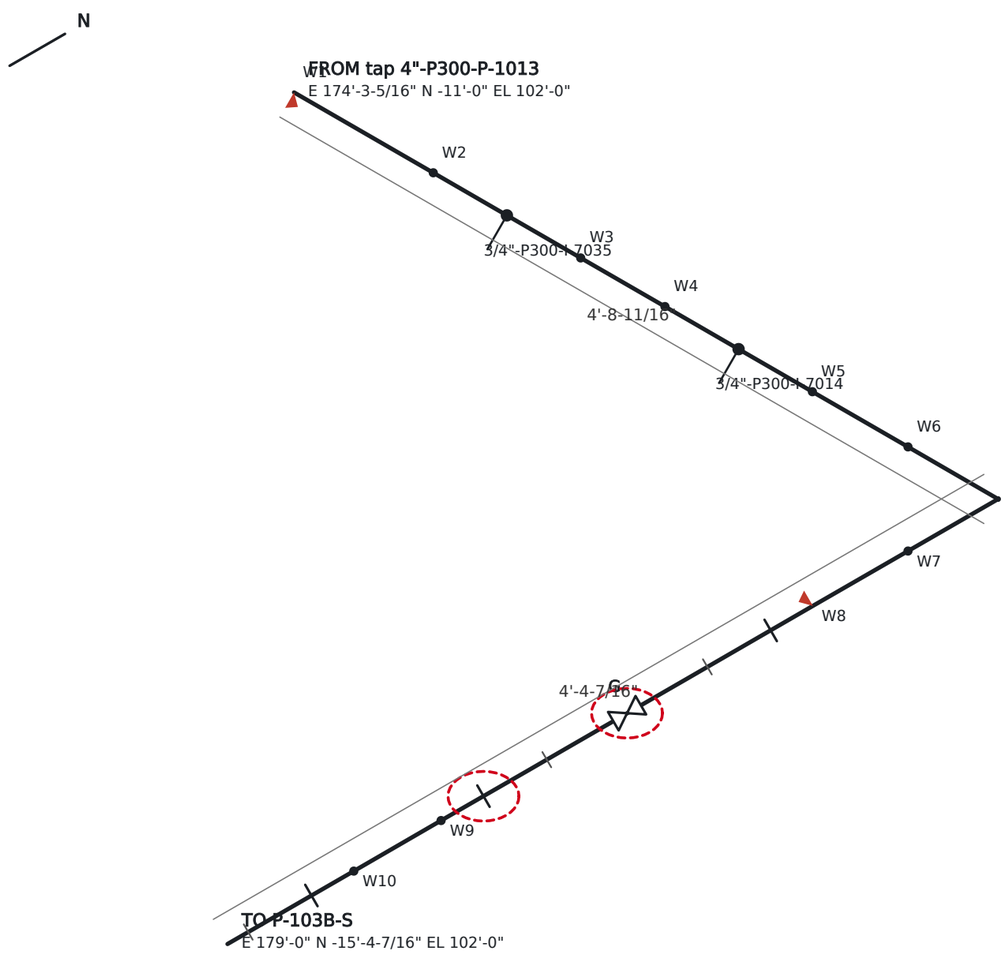
Description	Running at duty: no seal leak; bearing temperature and vibration within the owner's limits
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	As-found alignment recorded	Owner alignment procedure	Recorded	Alignment record	W	R	—				
3	Seal and bearings replaced; impeller and wear rings inspected	Owner pump procedure	Clearances within the owner's criteria	Pump overhaul record	S	R	—				
4	Nozzle joints remade with new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
5	Final alignment	Owner alignment procedure	Within the owner's alignment criteria	Alignment record	W	W	—				
6	Run test at duty	SW-TEST-2	No seal leak; temperatures and vibration within owner limits	Run test record	W	H	—				
7	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
8	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1014 — Overhaul pump P-103B (mechanical seal and bearings)



BILL OF MATERIALS

1	PIPE 4" P300 BEVELLED ENDS	4'-2-1/2"
2	TEE REDUCING 4" × 3/4" P300	2
3	ELBOW 90° R = 5/3 OD 4" P300	1
4	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	3
5	FLANGE WELD-NECK 4" P300 RAISED FACE	3
6	GASKET 4" P300 COMPRESSED SHEET	3
7	VALVE, GATE, WEDGE, OS&Y 4" P300	1

WELDS 10 (2 FIELD) · SUPPORTS 0 · SPOOLS 3

W1F W2 W3 W4 W5 W6 W7 W8F W9 W10

4"-P300-P-1014

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
tap 4"-P300-P-1013 → P-103B-S
PIPE 4'-2-1/2" · WEIGHT 287 lb

DWG ISO-4"-P300-P-1014 REV 0 WO-400-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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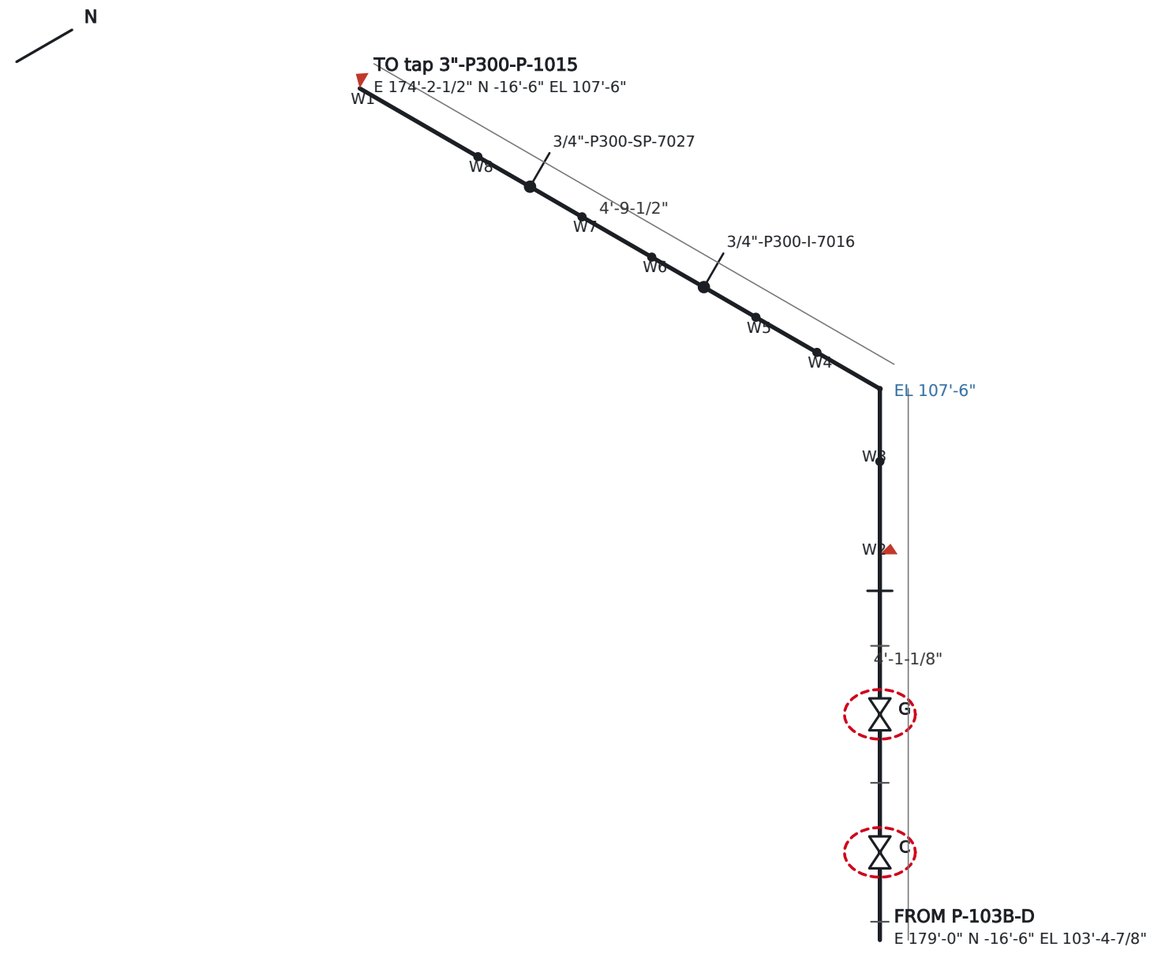
ISOMETRIC — 3"-P300-P-1016 — Overhaul pump P-103B (mechanical seal and bearings)



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	3
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	3
4	VALVE, GATE, WEDGE, OS&Y 3" P300	1
5	FLANGE WELD-NECK 3" P300 RAISED FACE	1
6	PIPE 3" P300 BEVELLED ENDS	4'-5-5/16"
7	ELBOW 90° R = 5/3 OD 3" P300	1
8	TEE REDUCING 3" × 3/4" P300	2

WELDS 8 (2 FIELD) · SUPPORTS 0 · SPOOLS 2

W1F W2F W3 W4 W5 W6 W7 W8

3"-P300-P-1016

CLASS H300 — Hydrocarbon process
SERVICE P · 3" · P300 · DESIGN 300 psi
P-103B-D → tap 3"-P300-P-1015
PIPE 4'-5-5/16" · WEIGHT 320 lb

DWG ISO-3"-P300-P-1016 REV 0 WO-400-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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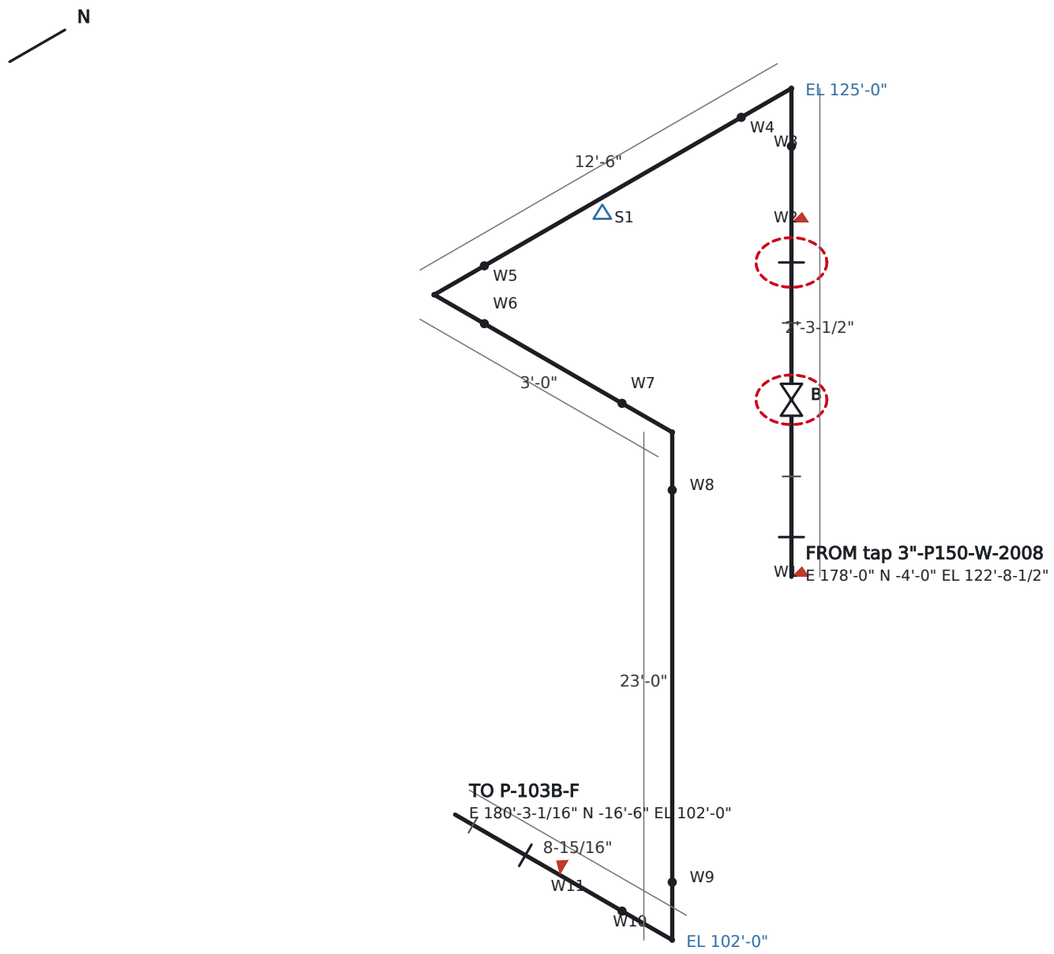
ISOMETRIC — 3/4"-P150-W-3022 — Overhaul pump P-103B (mechanical seal and bearings)



BILL OF MATERIALS

1	FLANGE WELD-NECK 3/4" P150 RAISED FACE	3
2	GASKET 3/4" P150 COMPRESSED SHEET	3
3	VALVE, BALL, FLOATING 3/4" P150	1
4	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150	3
5	PIPE 3/4" P150 BEVELLED ENDS	38'-5-7/16"
6	ELBOW 90° R = 5/3 OD 3/4" P150	4

WELDS 11 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

3/4"-P150-W-3022

CLASS W150 — Water (cooling and process)
SERVICE W · 0.75" · P150 · DESIGN 150 psi
tap 3"-P150-W-2008 → P-103B-F
PIPE 38'-5-7/16" · WEIGHT 68 lb

DWG ISO-3/4"-P150-W-3022 REV 0 WO-400-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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